POLITECNICO
MILANO 1863

Control systems

Robotics Academy

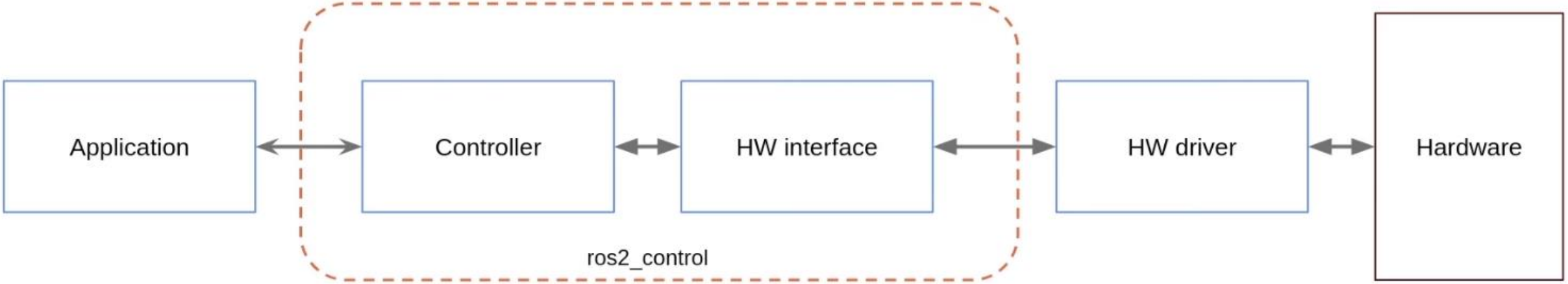
30/03/2026 | Lorenzo Ortolani, Gabriel Voss

Control systems in ROS2

02

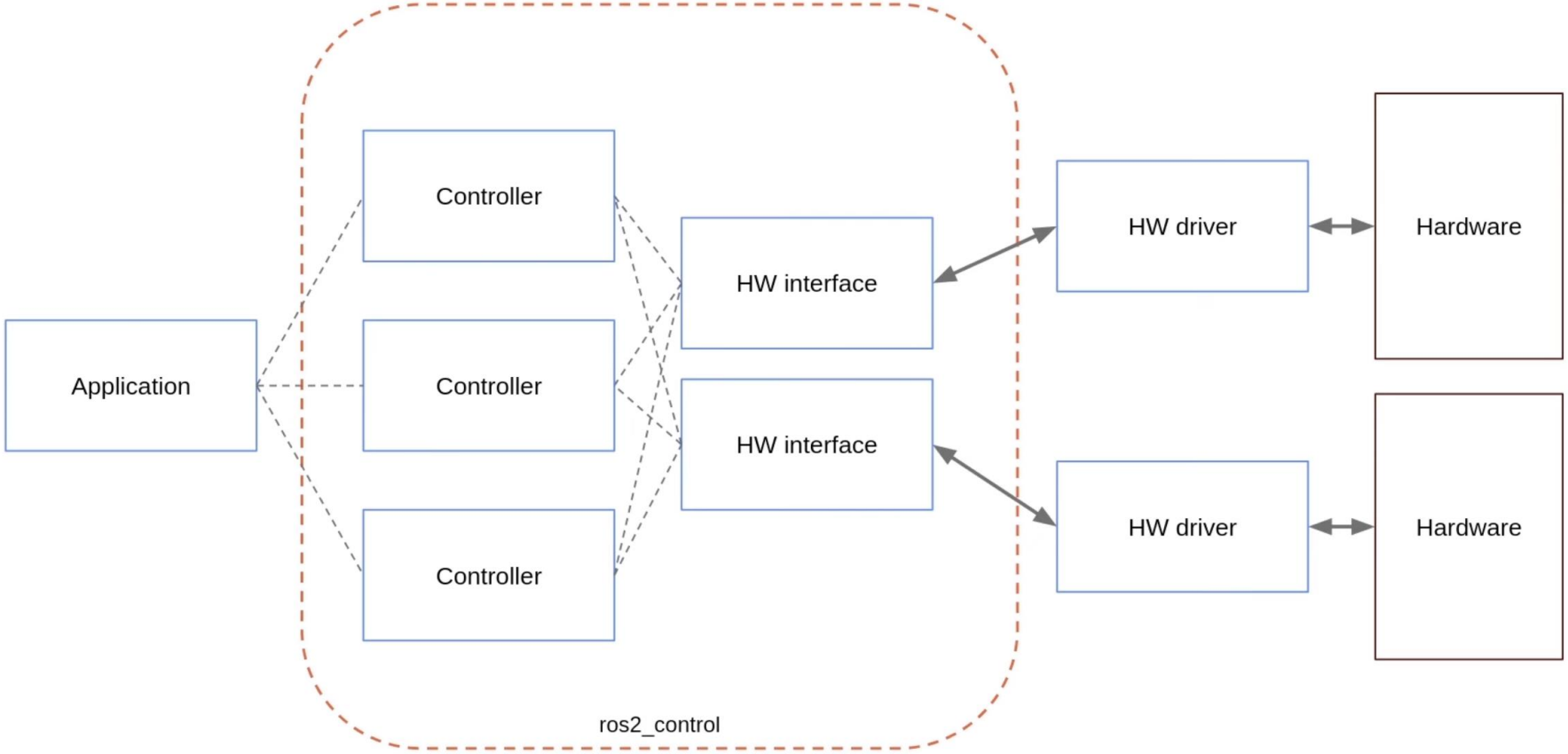
How does ros2_control works

One controller with one hardware interface (e.g. motion trajectory tracking controller of a mobile robot).



How does ros2_control works

More controllers with more hardware interfaces (e.g. controlling simultaneously two robotic arms for coordination tasks)

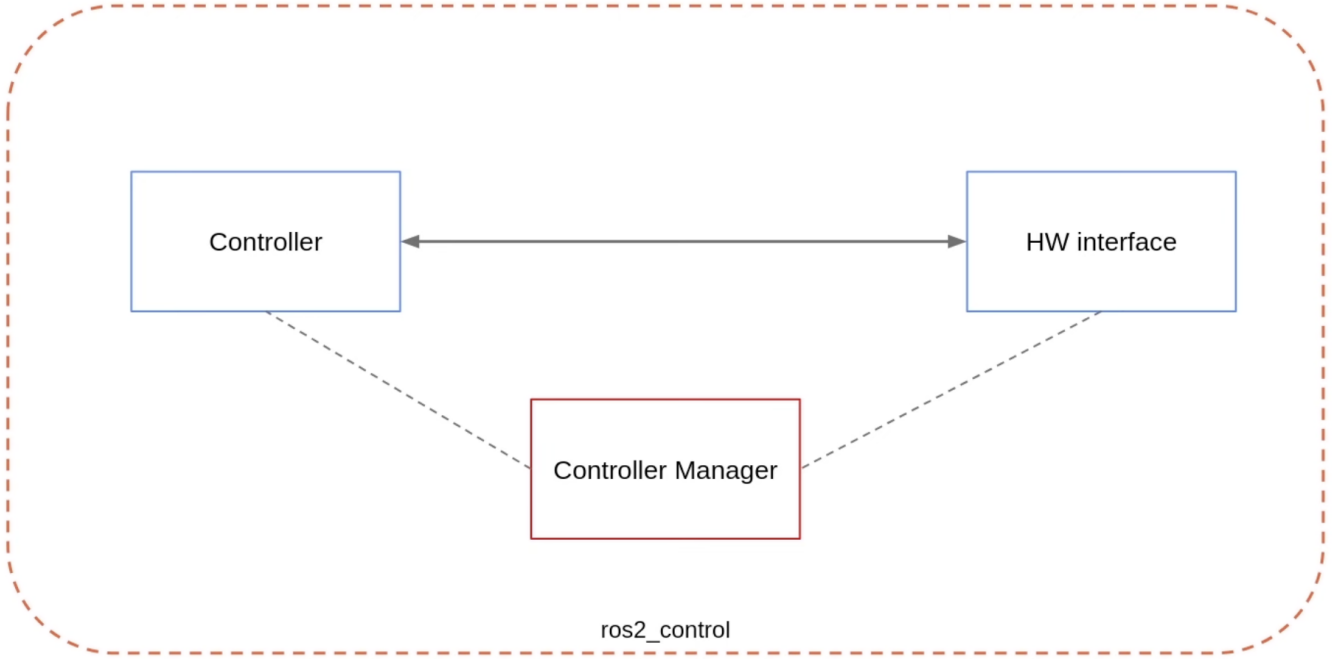


Controller Manager

Controller Manager is the main component in the ros2_control framework. It manages lifecycle of controllers, access to the hardware interfaces and offers services to the ROS-world.

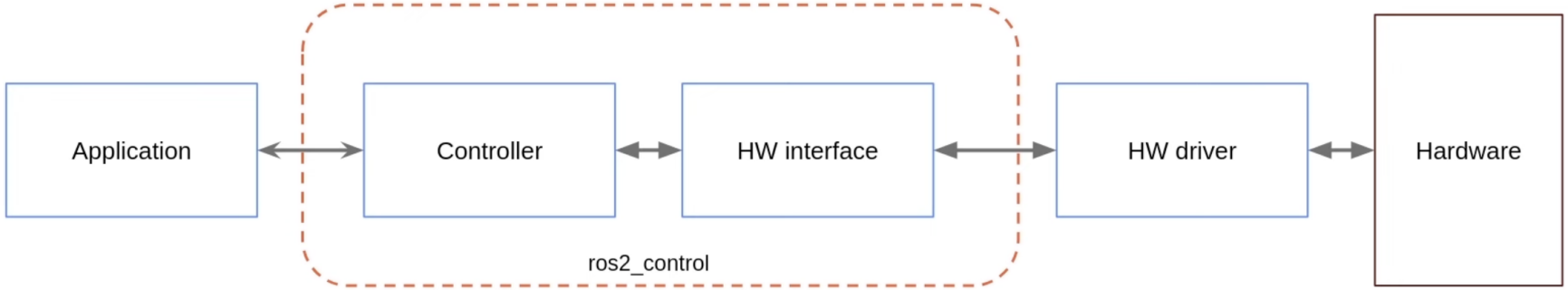
Key roles:

- Manages the lifecycle of controllers (load, configure, start, stop, unload).
- Provides access to hardware interfaces (sensors and actuators).
- Runs the main real-time control loop (reads hardware state -> updates controllers -> writes commands to hardware).



Our first example of ros2_control

We will implement a control system for a differential drive robot through ros2_control and our aim is to send reference velocities commands with a high level controller. The lower level controller will convert the control actions into reference velocities for each wheel.



→ /cmd_vel topic for the robot

- Linear/angular velocity
- m/s and rad/s

→ Transforms into a velocity for each wheel

- Angular velocity
- rad/s

→ Connection between controller and driver

→ Transforms cmd into something the motor can understand

- Ex: RPM